IITB-OPTIVER
AI INNOVATION LAB

Sponsored by Optiver

Presents
Perceptron Hackathon 2026

Introduction & Vision

The IITB-OPtiver AI Innovation Lab Hackathon invites builders to tackle ten concrete, technically demanding problems spanning efficient AI systems, causal reasoning, financial markets, multi-agent coordination, education, and trustworthy AI. Unlike a single open-ended brief, each theme below bundles multiple related problem statements. Teams pick one theme, then one specific problem within it, and are evaluated on the depth and rigor of their technical plan, not a working prototype, at this stage.

Competition Themes

Theme 1: Efficient AI at Scale

Problem 1: Cheap Fine-Tuning Across Weak, Scattered Devices

Context: Organizations often want to specialize a shared model using data that lives on many different machines without moving the raw data anywhere.

Challenge: Build a lightweight fine-tuning scheme where even low-power devices can meaningfully contribute updates to a shared model, training stays stable even when devices drop out mid-run, and one data-rich device can't quietly dominate the result.

Starter Resources: HuggingFace PEFT (LoRA) + an open federated learning framework (e.g. Flower), applied to a small open model (GPT-2 or

TinyLlama scale)

Minimum Viable:

- Simulate 3–5 nodes with different data splits and compute budgets
- Show the shared model converges despite the heterogeneity

Stretch Goals:

- Simulate node dropout mid-training
- Compare a smarter aggregation strategy against naive averaging

Problem 2: Squeezing Long-Context Inference into Less Memory

Context: Long documents or chat histories strain GPU memory, mostly from the cache a model keeps for past tokens.

Challenge: Shrink that cache significantly while barely hurting the model's ability to recall earlier content.

Starter Resources: Open reference implementations of existing cache-compression techniques (e.g. StreamingLLM, H2O), applied to a small model (1-3B params)

Minimum Viable:

- Apply an existing compression technique and measure memory savings vs. an uncompressed baseline
- Show the accuracy drop on a small, long-context recall task

Stretch Goals:

- Propose a modification to an existing method
- Benchmark against 2+ baselines

Note: Choose either Problem 1 or Problem 2 (not both) as your team's single problem statement for Round 1.

Theme 2: Cause and Effect in AI Decisions

Problem 3: Did the Policy Actually Work, or Did It Just Look Like It Did?

Context: A deployed decision system shapes the very outcomes later used to judge it,so simple before/after metrics can mislead.

Challenge: Estimate what would have happened under a different policy, using only logged data from the current one.

Starter Resources: A public bandit-feedback or off-policy evaluation dataset (e.g.

Open Bandit Dataset), or a synthetic dataset with known ground-truth counterfactuals.

Minimum Viable:

- Apply a standard off-policy evaluation method (e.g. inverse propensity weighting or a doubly robust estimator)
- Show a concrete case where this estimate differs from a naive before/after number

Stretch Goals:

- Quantify uncertainty on the counterfactual estimate
- A basic sensitivity analysis for unobserved confounding

Note: This theme currently has a single problem statement.

Theme 3: Markets & Trading Intelligence

Problem 4: Order Book Forecasting Under Regime Shifts

Context: Limit order book dynamics change abruptly around macro announcements, volatility spikes, and liquidity shocks. Models trained on one regime often degrade silently when the regime changes.

Challenge: Build a short-horizon LOB forecasting model that detects when the data distribution has shifted and adapts, instead of silently degrading.
Starter Resources: Public LOB datasets (e.g. LOBSTER sample data, or crypto exchange order-book data via exchange APIs)

Minimum Viable:

- Predict short-term price direction or spread from order book features
- Detect at least one distribution shift using a simple statistical test (e.g. rolling KS test) and show the model reacting to it

Stretch Goals:

- Multiple regime-adaptive prediction heads
- Latency benchmarking under a defined budget

Problem 5: Market-Impact-Aware Execution Strategy

Context: A firm's own large orders move the market it's trying to trade in, so naive models overstate expected returns.

Challenge: Build an execution agent that reduces slippage compared to a standard baseline by accounting for its own market impact.

Starter Resources: An open market simulator (e.g. ABIDES)

Minimum Viable:

- Implement a TWAP/VWAP baseline
- Build one agent (rule-based, heuristic, or simple RL) that measurably beats the baseline on slippage in the simulator

Stretch Goals:

- Explicit counterfactual modeling of "what the book would look like without our order"
- Multi-step feedback-loop analysis

Problem 6: Cross-Asset Statistical Arbitrage with Latency Budgets

Context: Arbitrage opportunities across correlated instruments exist for only milliseconds before they close.

Challenge:Identify and rank cross-asset mispricings in near real time, trading off Signal confidence against acting-on-it latency.

Starter Resources: Synthetic paired price series, or public multi-exchange crypto price data for the same asset.

Minimum Viable:

- Detect mispricings using a threshold or statistical model
- Report how fast the opportunity typically decays

Stretch Goals:

- Simulated end-to-end latency budget
- A basic paper-trading demo

Note: Choose exactly one of Problems 4, 5, or 6 as your team's problem statement.

Theme 4: Agents That Compete, Cooperate, and Adapt

Problem 7: Splitting a Shared Resource When Demand Keeps Changing

Context: Delivery fleets, compute clusters, or robot swarms must divide limited resources among agents while demand shifts unpredictably.

Challenge: Build a multi-agent system that reallocates resources as demand shifts, using some form of look-ahead rather than a fixed static policy.

Starter Resources: An open multi-agent simulation environment (e.g. a grid-world dispatch simulator, or PettingZoo-based environment) provided or pointed to by organizers.

Minimum Viable:

- Implement a rule-based baseline allocator

- Build one agent (RL or planning-based) that outperforms it when demand suddenly shifts

Stretch Goals:

- Handle both an abrupt shock and a gradual drift in the same run
- Evaluate over long horizons, not just short episodes

Note: This theme currently has a single problem statement.

Theme 5: AI That Teaches

Problem 8: A Tutor That Finds the Real Reason You're Stuck

Context: Struggling with quant reasoning, math, or code usually isn't about missing facts it's one specific misunderstanding producing the same class of mistake repeatedly.

Challenge: Build a tutoring system that traces a learner's wrong answers back to a likely root misunderstanding and generates targeted follow-up problems.

Starter Resources: A public dataset of student errors (e.g. ASSISTments, or a coding-error dataset), or a small hand-built taxonomy of 3–5 common misconceptions if no dataset fits.

Minimum Viable:

- Classify a wrong answer into one of 3–5 misconception categories
- Automatically generate a new practice problem targeted at that category

Stretch Goals:

- Track a learner's misconceptions across multiple sessions
- Show some evidence the targeted problems actually reduce that error type over time

Note: This theme currently has a single problem statement.

Theme 6: AI You Can Actually Trust

Problem 9: Catching an Agent Before It Acts on a Bad Idea

Context: Agents that call tools and take multi-step actions can hallucinate a plausible but ungrounded action.

Challenge: Build a guardrail layer that checks a proposed action against a verifiable source before it executes.
Starter Resources: An existing agent framework (LangChain/LangGraph or similar) plus a small reference knowledge base for grounding checks.
Minimum Viable:

- Build a checker that verifies at least one class of agent action against a source before execution
- Demonstrate it catching at least one deliberately bad/hallucinated action

Stretch Goals:

- Add routing between a fast/cheap model and a slower/careful one based on risk
- A small red-teaming test suite

Problem 10: Trusting a Signal Extracted from Messy, Manipulated Text

Context: Decisions increasingly lean on signals from unstructured text alongside numeric data, but that text is noisy and sometimes manipulated.
Challenge: Fuse text and time-series data into one signal and report a confidence bound on each prediction that's meaningful.
Starter Resources: A public financial news + price dataset (widely available on Kaggle),plus an existing conformal prediction library (e.g. MAPIE) instead of building calibration machinery from scratch.
Minimum Viable:

- Build a fusion model combining text sentiment and time-series signal
- Wrap predictions with an off-the-shelf conformal prediction method for confidence bounds

Stretch Goals:

- Test robustness under one adversarial or noisy-label scenario
- Check whether the confidence bounds are actually well-calibrated, not just present

Note: Choose either Problem 9 or Problem 10 as your team's problem statement.

Competition Structure & Timeline

Eligibility & Team Rules

- **Team Size:** A maximum of 3 members per team.

- **Eligibility:** Open to all students with a valid ID from their educational institution. Cross-institutional teams are encouraged.

Round Structure

Round 1: Online abstract submission, evaluated by a panel against the criteria below.

Round 2: Finals. A total of 25-30 teams are selected across all themes combined (not 30 per theme) to advance to the in-person finals at IIT Bombay.Teams must bring a working code / prototype demonstrating their Minimum Viable scope (and any stretch goals attempted).Round 1 was plan-only, but Round 2 requires an actual functioning build to present and demo to the judges.

Midway Check : Roughly 15 days before the finals, each of these shortlisted teams submits a 1-page progress report. This is a lightweight checkpoint to confirm the team is on track and has not stalled or pivoted away from its stated plan . it is not separately scored, but flags teams needing support or follow-up before the final judging.

Finals : At the finals, each team presents to a joint panel of judges from Optiver and IIT Bombay professors. Scores from the panel are normalized across all 30 teams (regardless of theme) on a common scale, and the top 3 teams overall (not top 3 per theme) are declared the winners.

Teams choose **one problem statement** from the **10 listed across the 6 themes** (Efficient AI at Scale, Cause & Effect in AI Decisions, Markets & Trading Intelligence, Agents, AI That Teaches, AI You Can Trust).

Abstract Format

Suggested length: 600–900 words (roughly 1–1.5 pages), plus an optional 1-page architecture/approach diagram.

Section	What to include
Problem & Theme chosen	Which of the 10 problem statements, and 2–3 sentences on why it's interesting/relevant to the team.
Understanding of the challenge	Restate the core technical difficulty in your own words — shows you've actually grasped what's hard about it, not just the surface prompt.
Proposed approach	Concrete method/architecture you intend to build. Name specific techniques, models, datasets, or tools. Avoid vague phrases like "we will use AI/ML to solve this."
Minimum Viable	Map your plan explicitly to the problem's Minimum Viable bullets ,what will exist and be demoable by the deadline.
Stretch ambitions	Which stretch goal(s) you'd attempt if time allows, and why they're the natural next step.
Novelty / differentiation	If your approach deviates from the "obvious" baseline, call it out explicitly.
Evaluation plan	How you'll show your solution actually works — metrics, baselines you'll compare against, datasets/simulators you'll use.
Feasibility and Risks	Brief note on relevant skills across the 3 members and why you can execute this in the available time. Also add what could go wrong?

Not required: code, results, or working prototypes at this stage. Round 1 is judged on the plan, not execution. (Though a rough diagram or pseudo-architecture is a strong plus.)

Evaluation Criteria

Criterion	Weight	What reviewers are checking for
Problem Understanding & Relevance	20%	Does the abstract correctly identify the actual technical crux of the problem? Is the scope realistic for the problem's Minimum Viable bar?
Technical Soundness & Feasibility	25%	Is the proposed method technically valid? Are named tools/datasets/techniques appropriate and actually suited to the problem? Can this be built by 3 people in the hackathon timeframe?
Innovation / Novelty	20%	Does the approach go beyond the obvious baseline named in "Starter Resources"? Any original ideas in method, evaluation, or framing?
Clarity & Structure of Abstract	10%	Is the writing precise and specific (concrete techniques/metrics) rather than buzzword-heavy? Can a reviewer picture the actual system being built?
Impact & Rigor of Evaluation Plan	15%	Are the proposed metrics/baselines meaningful? Would the comparison actually prove the claim
Team Fit & Readiness	10%	Does the team's stated skill mix plausibly support the plan? Is there a credible week-by-week or milestone sense of how they'll get to MVP?

Scoring scale (per criterion)

- 5 – Excellent: Precise, technically deep, clearly feasible, shows original thinking.
- 4–Strong: Solid plan, minor gaps in depth or specificity.

- 3–Adequate:Understands the problem but approach is generic/under-specified.
- 2–Weak:Misunderstands part of the problem, or approach is infeasible/vague.
- 1 – Poor: Boilerplate/buzzword abstract with no concrete technical plan.

Weighted score = Σ (criterion score × weight). Normalize to rank teams.

Final Presentations

Final presentations will be held at Techfest, IIT Bombay, from **December 15-18, 2026**. This is the stage where the top teams will compete for the final prizes. Please note that the decisions of the organizers and judges are final and binding

Submission Guidelines for Round 1 and Midway Report :
All online submissions for this competition must be made through the official Techfest website prior to the specified deadline. Only the Team Leader is authorized to submit on behalf of the team. Submissions made by any other team member will not be considered valid, multiple submissions from the same team will not be accepted and may lead to disqualification from the competition.

Key Dates

Event	Dates
Registration closes	15th October
Round 1 submission Deadline	31st October
Shortlist announcement	10th November
Midway Check	1st December
Finals	15-18th December

General Rules

1. Teams must register online at **techfest.org**, receiving a Team ID for future reference.
2. The organizers and judges' decisions are final and binding. Techfest reserves the right to verify participant details.
3. Techfest, IIT Bombay, is not responsible for late or lost entries.

4. Solutions must be original and not protected by patents, copyrights, or publications.
5. The latest information will be available on **techfest.org**, with registered participants being notified of changes via email.
6. All technical, procedural, and competition-related queries must be submitted directly through the Techfest Competition Portal under the Doubt section. Participants are strictly requested to regularly check the portal for answers and updates
7. Official communication for important announcements will be through the email satwik@techfest.org. Queries submitted via email will not be entertained.

Certificate Policy

1. The top three teams in Round 2 receive a **Certificate of Excellence**.
2. All the teams that are qualified for round 2 will receive an **E-certificate**

Prize Money

Prize money will be awarded to the **top three teams** via NEFT, latest by May 2027.

Prize Money Submission Process

- **Submission Channel:** Complete the prize money form **directly on the website** after the final round concludes. The winners would be able to view the form through their registered email ID
- **Deadline:** The form must be submitted by **December 31, 2026**.

Details Needed for the Website Form

When filling out the form on the website, make sure you have the following bank account details ready:

1. Account Holder's Name
2. Account Number
3. Bank Name and Branch Name
4. IFSC Code